

1. Analysis of Chemical Composition in Living Organisms

• Elemental Analysis:

- Performing elemental analysis on plant tissue, animal tissue, or microbial paste yields a list of elements like carbon, hydrogen, oxygen, and others, with their content per unit mass.
- Comparing this to a non-living sample (e.g., earth's crust), a similar list of elements is obtained, indicating no absolute differences in the presence of elements.
- However, a closer examination reveals that the relative abundance of carbon and hydrogen is higher in living organisms compared to the earth's crust.
- Elemental analysis provides the elemental composition (e.g., hydrogen, oxygen, chlorine, carbon).

• Chemical Analysis for Compounds:

- To find organic compounds, a living tissue can be ground in trichloroacetic acid (Cl_3CCOOH) using a mortar and pestle to obtain a thick slurry.
- Straining this slurry through cheesecloth or cotton yields two fractions:
 - The filtrate, or acid-soluble pool, contains thousands of organic compounds.
 - The retentate, or acid-insoluble fraction.
- Compounds are extracted, then subjected to various separation techniques to isolate and purify them.
- Analytical techniques then determine their molecular formula and probable structure.
- All carbon compounds obtained from living tissues are called biomolecules.
- Analysis for compounds reveals the types of organic (Figure 9.1) and inorganic (Table 9.2) constituents present in living tissues.

• Detecting Inorganic Elements and Compounds:

- A destructive experiment is required to find inorganic elements and compounds.
- Weighing a living tissue (wet weight) and drying it removes all water, leaving the dry weight.
- Burning the dried tissue oxidizes all carbon compounds to gaseous forms (CO_2 , water vapor), which are removed.
- The remaining material is called 'ash', which contains inorganic elements like calcium, magnesium, etc..
- Inorganic compounds such as sulphate and phosphate are also found in the acid-soluble fraction.

11. Classification of Biomolecules

• Micromolecules (Biomolecules):

- Compounds found in the acid-soluble pool.
- Have molecular weights ranging from 18 to around 800 daltons (Da) approximately.
- Examples include amino acids, monosaccharide and disaccharide sugars, fatty acids, glycerol, nucleotides, nucleosides, and nitrogen bases.

• Macromolecules (Biomacromolecules):

- Compounds found in the acid-insoluble fraction.
- With the exception of lipids, these compounds have molecular weights in the range of ten thousand daltons and above.
- These are polymeric substances, except for lipids.
- Four types of organic compounds: proteins, nucleic acids, polysaccharides, and lipids.
- Lipids are an exception: Despite having molecular weights not exceeding 800 Da (small molecular weight compounds), they appear in the acid-insoluble fraction. This is because lipids are arranged into structures like cell membranes. When tissue is ground, cell membranes break into non-water-soluble

vesicles that separate with the acid-insoluble pool. Therefore, lipids are not strictly macromolecules.

• Primary vs. Secondary Metabolites:

- Primary Metabolites: Biomolecules like amino acids and sugars found in animal tissues. They have identifiable functions and play known roles in normal physiological processes.
- Secondary Metabolites: Thousands of compounds found in plant, fungal, and microbial cells that are other than primary metabolites. Examples include alkaloids, flavonoids, rubber, essential oils, antibiotics, pigments, scents, gums, and spices (Table 9.3). Their role or function in host organisms is not fully understood. However, many are useful for human welfare (e.g., rubber, drugs, spices, scents, pigments) or have ecological importance.

III. Types of Biomolecules

1. Amino Acids:

- Organic compounds containing an amino group ($-NH_2$) and an acidic (carboxyl) group ($-COOH$) attached to the same carbon (α -carbon).
- They are "substituted methanes" with four substituent groups: hydrogen, carboxyl group, amino group, and a variable R group.
- There are twenty types of amino acids that occur in proteins. The R group defines the amino acid (e.g., hydrogen in glycine, methyl in alanine, hydroxy methyl in serine).
- Their chemical and physical properties are determined by the amino, carboxyl, and R functional groups.
- Classification based on number of groups:
 - Acidic: e.g., glutamic acid.
 - Basic: e.g., lysine.
 - Neutral: e.g., valine.
- Aromatic amino acids: e.g., tyrosine, phenylalanine,

tryptophan.

- Amino acids have an ionizable nature of their $-NH_2$ and $-COOH$ groups, so their structure changes with pH (e.g., zwitterionic form).
- Amino acids can be essential (must be supplied through diet) or non-essential (body can make them).

2. Lipids:

- Generally water insoluble.
- Fatty acids: Have a carboxyl group attached to an R group, which can be a methyl, ethyl, or longer chain of $-CH_2$ groups (1 to 19 carbons).
 - Palmitic acid has 16 carbons (including carboxyl carbon).
 - Arachidonic acid has 20 carbons (including carboxyl carbon).
 - Can be saturated (no double bonds) or unsaturated (one or more $C=C$ double bonds).
- Glycerol: Trihydroxy propane.
- Glycerides: Many lipids have both glycerol and fatty acids, where fatty acids are esterified with glycerol. They can be:
 - Monoglycerides.
 - Diglycerides.
 - Triglycerides.
 - Also called fats and oils based on melting point. Oils have lower melting points (e.g., gingelly oil) and remain liquid in winter.
- Phospholipids: Contain phosphorous and a phosphorylated organic compound. They are found in cell membranes (e.g., lecithin).
- Some lipids, especially in neural tissues, have more complex structures.

3. Nitrogen Bases, Nucleosides, and Nucleotides:

- Nitrogen bases: Heterocyclic rings containing nitrogen. Examples: adenine, guanine, cytosine, uracil, and thymine.

- Adenine and Guanine are purines.
- Uracil, Cytosine, and Thymine are pyrimidines.
- **Nucleosides:** Nitrogen bases attached to a sugar. Examples: Adenosine (adenine + sugar), guanosine, thymidine, uridine, cytidine.
- **Nucleotides:** A nucleoside with a phosphate group esterified to the sugar. Examples: Adenylic acid, thymidylic acid, guanylic acid, uridylic acid, cytidylic acid. A nucleotide has three components: a heterocyclic compound (nitrogenous base), a monosaccharide (sugar), and a phosphoric acid/phosphate.
- **Sugars in nucleotides:** Either ribose (a pentose monosaccharide) or 2'-deoxyribose.
- **Nucleic Acids:** Polymers consisting only of nucleotides.
- Deoxyribonucleic acid (DNA): Contains deoxyribose sugar.
- Ribonucleic acid (RNA): Contains ribose sugar.
- Both DNA and RNA function as genetic material and carry hereditary information.

4. Proteins:

- **Polypeptides:** Linear chains of amino acids linked by peptide bonds.
- **Heteropolymers:** Made of different building blocks; a protein is a polymer of amino acids.
- **Twenty types of amino acids** serve as monomers.
- **Functions:** Carry out many vital functions in living organisms.
- Transport nutrients across cell membranes.
- Fight infectious organisms (e.g., antibodies).
- Hormones (e.g., insulin).
- Enzymes (e.g., trypsin, carbonic anhydrase).
- Receptors for sensory reception (smell, taste, hormones).
- Structural proteins (e.g., collagen, intercellular ground substance).

- GLUT-4 enables glucose transport into cells.

- **Abundance:**

- Collagen is the most abundant protein in the animal world.
- Ribulose biphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in the entire biosphere.
- Average composition in cells: Proteins constitute 10-15% of total cellular mass.

5. Polysaccharides (Carbohydrates):

- **Macromolecules:** Long chains of sugars.
- **Threads** containing different monosaccharides as building blocks.
- **Homopolymers:** Consisting of only one type of monosaccharide.
- **Cellulose:** Polymeric polysaccharide made only of glucose. Plant cell walls are made of cellulose; paper and cotton are cellulosic. Cellulose does not contain complex helices and cannot hold 12.
- **Starch:** A variant of cellulose, acts as an energy storehouse in plant tissues. Starch forms helical secondary structures and can hold 12 molecules, resulting in a blue color.
- **Glycogen:** An energy storage variant in animals.
- **Inulin:** A polymer of fructose.
- In a polysaccharide chain (e.g., glycogen), the right end is the reducing end, and the left end is the non-reducing end.
- **Complex Polysaccharides:** Have amino-sugars and chemically modified sugars as building blocks (e.g., glucosamine, N-acetyl galactosamine).
- **Chitin:** A complex polysaccharide found in the exoskeletons of arthropods. Most complex polysaccharides are homopolymers.
- Average composition in cells: Carbohydrates constitute 3% of total cellular mass.

IV. Structure of Proteins

Biologists describe protein structure at four levels:

1. Primary Structure:

- The sequence of amino acids (positional information) in a protein.
- A protein is imagined as a line; the left end is the N-terminal amino acid (first), and the right end is the C-terminal amino acid (last).

2. Secondary Structure:

- The protein thread folds into a helix (like a revolving staircase).
- Only right-handed helices are observed in proteins.
- Other regions of the protein thread fold into other forms, such as beta-pleated sheets (Figure 9.3 b).

3. Tertiary Structure:

- The long protein chain folds upon itself like a hollow woolen ball, creating a 3-dimensional view.
- This structure is absolutely necessary for the many biological activities of proteins. When looking at the tertiary structure, the chain criss-crosses, forming crevices or pockets. One such pocket is the 'active site'.

4. Quaternary Structure:

- An assembly of more than one polypeptide or subunits.
- Describes the manner in which these individual folded polypeptides or subunits are arranged relative to each other.
- Example: Adult human haemoglobin (Hb) consists of 4 subunits (two α -type and two β -type).

V. Enzymes

• Nature and Composition:

- Almost all enzymes are proteins.
- Some nucleic acids also behave like enzymes, called ribozymes.
- Like other proteins, enzymes possess primary, secondary, and

tertiary structures.

- Their tertiary structure creates crevices or pockets, one of which is the 'active site'. An active site is where the substrate fits.

• Catalytic Action:

- Enzymes catalyse reactions at a high rate through their active site.

- They dramatically accelerate reaction rates compared to uncatalyzed reactions. For example, carbonic anhydrase increases the formation of H_2CO_3 by about 10 million times.

- The chemical converted into a product is called the 'substrate' (S). Enzymes convert S into a product (P).

○ Mechanism (Catalytic Cycle):

1. The substrate (S) binds to the active site of the enzyme (E), forming a highly reactive, short-lived enzyme-substrate complex (ES). This complex formation is essential for catalysis.

2. The binding of the substrate induces the enzyme to alter its shape, fitting more tightly around the substrate.

3. Within the ES complex, a transition state structure of the substrate is formed, which is an unstable intermediate.

4. The active site, in close proximity to the substrate, breaks and forms chemical bonds of the substrate, leading to the formation of a new enzyme-product complex (EP).

5. The enzyme releases the products (P) from the active site, and the free enzyme is ready to bind to another substrate molecule, repeating the cycle.

• Activation Energy:

- Chemical reactions require substrates to pass through a much higher energy state or transition state.

- The difference in average energy content between the substrate and this transition state is called 'activation energy'.

- Enzymes bring down this energy barrier (activation energy), making the transition from substrate to product easier.

- Chemical Reactions and Metabolic Pathways:

- A chemical reaction involves breaking and forming new bonds during transformation (e.g., hydrolysis of starch to glucose).

- The rate of a process is the amount of product formed per unit time. Rates are influenced by temperature.

- A metabolic pathway is a multistep chemical reaction where each step is catalyzed by the same or different enzymes.

- Example: Glucose becoming pyruvic acid through ten different enzyme-catalyzed reactions. This pathway can lead to various end products depending on conditions (e.g., lactic acid in anaerobic skeletal muscle, pyruvic acid in aerobic conditions, ethanol in yeast fermentation).

- Factors Affecting Enzyme Activity:

- The activity of an enzyme is affected by conditions that alter its tertiary protein structure.

- Temperature:

- Each enzyme has an optimum temperature at which it shows highest activity.

- Activity declines both below and above the optimum.

- Low temperature preserves the enzyme in a temporarily inactive state.

- High temperature (above 40°C) destroys enzymatic activity because proteins are denatured by heat.

- However, enzymes from thermophilic organisms (living in high temperatures) are stable and retain catalytic power up to $80^{\circ}\text{--}90^{\circ}\text{C}$.

- pH:

- Each enzyme has an optimum pH for highest activity.

- Activity declines both below and above the optimum.

- **Concentration of Substrate:**

- As substrate concentration increases, the reaction velocity initially rises.
- It then reaches a maximum velocity (V_{max}), which is not exceeded by further increases in substrate concentration. This occurs because all enzyme molecules become saturated with substrate, and no free enzyme molecules are available to bind additional substrate.

- **Inhibitors:**

- Specific chemicals can bind to an enzyme and shut off its activity; this is called inhibition, and the chemical is an inhibitor.

- **Competitive inhibitors:** Closely resemble the substrate in molecular structure and compete with the substrate for the enzyme's substrate-binding site. This reduces enzyme action (e.g., inhibition of succinic dehydrogenase by malonate). Competitive inhibitors are used in bacterial pathogen control.

- **Classification and Nomenclature:**

- Thousands of enzymes have been discovered and classified into 6 classes, each with 4-13 subclasses, and named by a four-digit number.

- **Oxidoreductases/dehydrogenases:** Catalyze oxidation-reduction between two substrates (e.g., $S_{reduced} + S'_{oxidised} \rightarrow S_{oxidised} + S'_{reduced}$).

- **Transferases:** Catalyze the transfer of a group (other than hydrogen) between a pair of substrates (e.g., $S-G + S' \rightarrow S + S'-G$).

- **Hydrolases:** Catalyze hydrolysis of various bonds (ester, ether, peptide, glycosidic, C-C, C-halide, P-N).

- **Lyases:** Catalyze removal of groups from substrates by mechanisms other than hydrolysis, leaving double bonds.

- **Isomerases:** Catalyze inter-conversion of optical, geometric,

or positional isomers.

- **Ligases:** Catalyze the linking together of two compounds (e.g., joining C-O, C-S, C-N, P-O bonds).

- **Co-factors:**

- **Non-protein constituents** that bind to enzymes to make them catalytically active.

- The protein portion of the enzyme without the cofactor is called the **apoenzyme**.

- **Three kinds of cofactors:**

- **Prosthetic groups:** Organic compounds tightly bound to the apoenzyme (e.g., haem in peroxidase and catalase, part of the active site).

- **Co-enzymes:** Organic compounds with transient association to the apoenzyme, often serving as co-factors in multiple reactions. Essential chemical components of many coenzymes are vitamins (e.g., NAD and NADP contain niacin).

- **Metal ions:** Form coordination bonds with side chains at the active site and with the substrate (e.g., zinc in carboxypeptidase).

- **Catalytic activity is lost when the co-factor is removed, highlighting their crucial role.**

VI. Average Composition of Cells

- **Water:** 70-90% (most abundant chemical in living organisms).

- **Proteins:** 10-15%.

- **Nucleic acids:** 5-7%.

- **Carbohydrates:** 3%.

- **Lipids:** 2%.

- **Ions:** 1%.

- The acid-soluble pool roughly represents cytoplasmic composition, while macromolecules from cytoplasm and organelles form the acid-insoluble fraction. Together, they

represent the entire chemical composition of living tissues

